HW2 **CS 641: Modern Cryptography**

Due date: 10th April 2026. 11:59 pm.

Submission process: Submit (i) source codes in one C/Python file, (ii) the commands+output of your programs in a text file, and (iii) the answer to question 3 in a pdf file. If I suspect copying, I might ask you to appear in a viva at a very short notice. The filename of your submissions should start with your roll number. (e.g. if your roll number is 1234 then either of the following file names are acceptable: 1234.c, 1234_name.c, 1234.abcd.c etc.)

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1. (a) Compute the SHA256 hash of your email-id. Let it be h_1.
   (b) Find another IIT Kanpur email-id (i.e. the format of the id should be x@iitk.ac.in where you need to find a suitable value for the variable x) which has the hash value h_2, such that h_1 and h_2 match in the first 16 bits. How much can you increase this number? 32 bits, 40 bits, 64 bits? Find the longest match that you can obtain.
   (c) Repeat the above a few times (say, 20, 50 or 100 times) and store the number of calls you made to the SHA256 hashing algorithm in each of these trials. If the average number of calls to SHA256 is t then report the value of $\log_2 t$.
   (d) Comment on the value you reported above. Why are you getting this specific value?
   (e) Find two email-ids of the form x@iitk.ac.in and y@iitk.c.in such that their hash values match in the first 16 bits. How much can you increase this number? Find the longest match that you can obtain.
   (f) Repeat the above a few times, and report the average number of trials as in part (c).
   (g) Comment on the value you reported above. Why are you getting this value? Why is it different from part (c)?

2. For this question, implement long integers suitably. Note that you can't use, for instance, `int` in C programming language since it supports only 32 or 64 bit long values. In this question, you are operating on much larger integers. You are allowed to use "long int" libraries in whatever programming language you are using.
   (a) Implement the Miller-Rabin primality test. Note that you are not allowed to call a library for this. You should implement this yourself. (For modular arithmetic, and for square and multiply algorithm, you are allowed to call library functions). However, the primality testing algorithm should be implemented by you on your own.
   (b) Use part(a) to generate random primes p and q of size 512 bits each.
   (c) Implement Extended Euclidean Algorithm (EEA) to compute gcd and the linear combination for two (long) integers. (Again, no library calls for this algorithm; implement this on your own).
   (d) Set $n = p \cdot q$ and compute $d = e^{-1} \bmod \phi(n)$ by using the EEA that you implemented.
   (e) generate a random message of 1024 bits, encrypt it using the textbook RSA, decrypt the ciphertext, and show that you get the same message after decryption.

3. (a) If a cyclic group has n elements then show that it has $\phi(n)$ generators.

(b) From Part (a), we get that the multiplicative group Z_p^* has $\phi(p-1)$ generators. (Recall from the discussion in the class that for all primes p, Z_p^* is cyclic and contains $(p-1)$ elements.) Primes p such that $(p-1) = 2 \cdot p'$ for another prime p' are called Sophie-Germaine primes. Generate a random Sophie-Germain prime p of size 512 bits.

(c) Find a generator of the group Z_p^* for the prime you generated above by randomly picking elements from the group. (Test for generator: An element of the group is a generator if the order of this element is $(p-1)$.)

(d) Implement the Diffie-Hellman Key Exchange as studied in the class. That is, exchange g^a and g^b between two parties and show that both parties compute the common secret.

General comment Not to be submitted.

Read about Sophie-Germaine and her life. Wikipedia article on her is a nice summary. Being a woman mathematician, she was discouraged from studying Mathematics. To hide her identity, she wrote papers under a pseduonym and did correspondence with other mathematicians of her time (Gauss was one of them) by using the same psuedonym. Despite all the obstacles, she succeeded in doing amazing work. Find out more about her life and work. We recently celebrated Women's Day. This is our tribute to a great mathematician.